

Supplement: Execution Before Evaluation in LLM Social Simulations

Anonymous Shared-Task Submission

1 Scope and contribution

The main paper reports a five-study claim-identifiable audit over 5,213 released runs. This supplement contains the complete atomic-claim tables, execution projections, identifiability decisions, source provenance, and reproduction contract. A positive signed effect supports the registered direction. Target verdicts are S (supported), C (contradicted), M (mixed/inconclusive), and U (unidentified). Design verdicts concern only non-default cells in which treatment and reference arms can be matched; absence of such cells is not interpreted as stability.

2 Method

For a paired claim h , treatment T_h and reference R_h are matched at the same seed and every remaining signal variable. Its signed run effect is

$$\tau_h = s_h [m(T_h) - m(R_h)], \quad s_h \in \{-1, +1\},$$

where the sign is chosen so that $\tau_h > 0$ supports the registered hypothesis. One-arm hypotheses use $\tau_h = m - \theta_h$. Let c index matched context strata. The reported point estimate is $|C|^{-1} \sum_c \bar{\tau}_{h,c}$, so contexts with denser grids do not dominate. The interval is the envelope of two 95% sensitivity intervals: resample contexts then seeds within context, and resample the released seed labels. Bootstrap draws are computational replicates, not independent observations.

A design level d is identifiable for h only if both claim arms exist under d , every other design variable is at default, and the matching keys above remain available. We separately report: (i) the hypothesis direction in the non-default cell, (ii) the matched-default direction on the same context subset, and (iii) the incremental design shift. No global max/max ratio is used.

The execution gate has four interface-specific forms: exact replay of released free-text parsers; generated tool-call to event replay; source inspection of the pinned runtime’s batching and turn policy; and strict-format probes as negative controls. Recorded events remain authoritative. Final-field or one-decision projections are sensitivity analyses, not claims about latent model intent.

3 Released evidence

Table 1: Cross-study execution evidence. Difference has the interface-specific meaning in the final column.

Study	runs	interface	decisions/probes	events	difference	scientific impact
Beauty Contest	670	free-text integer action	1,340	1,340	230	target verdict changes: H2;H6b
Iterated PD	3,465	free-text categorical action	69,300	69,300	58	no target-verdict change; effect sensitivity retained
Persona Expression	528	multi-tool social post	63,360	65,649	2,289	max per-run diversity shift 0.0782; no selected target-verdict change
Observed Norms	330	strict integer probe	36,720	36,720	0	execution-clean; three human-anchor claims remain data-unidentifiable
Polarization	220	open-ended multi-tool turn plus integer probe	190,784	271,970	60,634	23,327 turns exceed cap; 20 generated calls rejected; 3 probes failed

The free-text game discrepancies total 288 actions. Beauty contributes 230/1,340, changing target verdicts for H2, H6b; IPD contributes 58/69,300 with no target-verdict change. Persona Expression executes 2,289 posts beyond its 63,360 model decisions; the one-decision/one-post projection changes per-run Jaccard diversity by at most 0.0782. In Polarization, 23,327 agent-turns exceed the three-action cap by 60,634 actual events; the largest executed turn contains 686 events. The same sweep has 20 generated calls without events and 3 failed probes. Observed Norms is the clean control: all 36,720 responses are exactly one in-range integer.

4 All atomic claims

Table 2: All auditable atomic hypotheses. Intervals are conditional on released seeds.

Study	ID	claim	effect	95% CI	target	identifiable-design verdict
Beauty	H1	basic: 17–19 over 19–20	-.250	[-.750, .000]	M	mixed across identifiable cells (8 levels)
Beauty	H2	cycle shifts choice upward	1.550	[.378, 2.791]	S	mixed across identifiable cells (4 levels)
Beauty	H3	costless majority at 17–20	.500	[.500, .500]	S	positive direction but uncertain (4 levels)
Beauty	H4	strategic cue lowers choice	-.033	[-.198, .133]	M	mixed across identifiable cells (7 levels)
Beauty	H5a	9B depth minus 4B	.675	[.150, 1.337]	S	mixed across identifiable cells (1 levels)
Beauty	H5b	27B depth minus 9B	-1.833	[-3.008, -.825]	C	negative direction but uncertain (1 levels)
Beauty	H6a	cautious raises choice	-.312	[-1.863, 1.108]	M	not identifiable (0 levels)
Beauty	H6b	competitive lowers choice	-1.971	[-3.392, -.600]	C	not identifiable (0 levels)
IPD	P1a	moralized raises cooperation	.067	[.015, .120]	S	mixed across identifiable cells (9 levels)
IPD	P1b	risk lowers cooperation	.252	[.181, .324]	S	positive direction all identifiable cells (9 levels)
IPD	P2a	attenuated stakes lower cooperation	.079	[.041, .123]	S	mixed across identifiable cells (9 levels)
IPD	P2b	amplified stakes do not lower cooperation	-.057	[-.105, -.014]	C	mixed across identifiable cells (9 levels)
IPD	P3	business exceeds fictional cooperation	.046	[.017, .081]	S	mixed across identifiable cells (9 levels)
IPD	P5	early cooperation exceeds final-round cooperation	.043	[.026, .062]	S	mixed across identifiable cells (9 levels)
IPD	P6a	cooperative stance raises cooperation	.304	[.221, .395]	S	not identifiable (0 levels)
IPD	P6b	competitive stance lowers cooperation	.322	[.243, .400]	S	not identifiable (0 levels)
IPD	P6c	reciprocal stance increases previous-opponent matching	.001	[.000, .002]	M	not identifiable (0 levels)
Persona	E1a	demographic exceeds generic diversity	.081	[.050, .112]	S	not identifiable (0 levels)
Persona	E1b	full biography exceeds demographic diversity	.004	[.002, .007]	S	not identifiable (0 levels)
Persona	E1c	sociopsychological exceeds full-biography diversity	-.012	[-.015, -.010]	C	not identifiable (0 levels)
Persona	E3a	4B diversity is at least 9B diversity	-.004	[-.024, .014]	M	not identifiable (0 levels)
Persona	E3b	9B diversity is at least 27B diversity	-.026	[-.046, -.007]	C	not identifiable (0 levels)
Norms	N3	observation reduces within-population response dispersion	.572	[.224, .965]	S	
Polarization	Z1	stance-similar exposure raises edge-alignment gain	-.032	[-.053, -.012]	C	nonpositive all identifiable cells (5 levels)

Study	ID	claim	effect	95% CI	target	identifiable-design verdict
Polarization	Z6	stance-similar exceeds opposite-stance clustering	-.003	[-.020, .015]	M	not identifiable (0 levels)
Polarization	Z2a	scale-free exceeds random under chronological exposure	.001	[-.024, .021]	M	not identifiable (0 levels)
Polarization	Z2b	small-world exceeds random under chronological exposure	-.036	[-.058, -.018]	C	not identifiable (0 levels)
Polarization	Z2c	scale-free exceeds random under stance-similar exposure	-.012	[-.033, .009]	M	not identifiable (0 levels)
Polarization	Z2d	small-world exceeds random under stance-similar exposure	-.030	[-.053, -.004]	C	not identifiable (0 levels)
Polarization	Z3	no-memory increases opinion volatility	-.023	[-.029, -.017]	C	not identifiable (0 levels)
Polarization	Z4	credulous personas shift more than skeptical personas	.170	[.114, .232]	S	positive direction all identifiable cells (5 levels)

4.1 Unavailable registered claims

Study	parent claim	verdict	reason
Observed Norms	h1 country direction	U	registered WVS human anchors are not distributed because of licensing
Observed Norms	h2 private homogenization	U	claim compares model dispersion with WVS respondent dispersion, which is not distributed
Observed Norms	h5 model capability	U	directional accuracy is defined relative to unavailable WVS anchors

5 Execution-projection sensitivity

Table 3: Target sensitivity to the alternative observable projection. For games, “explicit” is the prompt-specified final action field. For Persona, “strict” retains the first post from each model decision.

Study	claim	recorded/executed	verdict	alternative	verdict	changed?
Beauty	H1	-.250	M	.000	M	no
Beauty	H2	1.550	S	1.237	M	yes
Beauty	H3	.500	S	.500	S	no
Beauty	H4	-.033	M	-.033	M	no
Beauty	H5a	.675	S	.675	S	no
Beauty	H5b	-1.833	C	-1.833	C	no
Beauty	H6a	-.312	M	1.021	M	no
Beauty	H6b	-1.971	C	-1.637	M	yes
IPD	P1a	.067	S	.068	S	no
IPD	P1b	.252	S	.253	S	no
IPD	P2a	.079	S	.081	S	no
IPD	P2b	-.057	C	-.058	C	no
IPD	P3	.046	S	.047	S	no
IPD	P5	.043	S	.042	S	no
IPD	P6a	.304	S	.304	S	no
IPD	P6b	.322	S	.317	S	no
IPD	P6c	.001	M	.001	M	no
Persona	E1a	.081	S	.084	S	no
Persona	E1b	.004	S	.004	S	no
Persona	E1c	-.012	C	-.012	C	no
Persona	E3a	-.004	M	-.006	M	no
Persona	E3b	-.026	C	-.027	C	no

Only the Beauty H2 and H6b target verdicts change. This distinction matters: an execution discrepancy can be real without overturning a particular scientific conclusion. Conversely, agreement of a target verdict does not validate the interface; it only bounds the consequence for that claim and released sweep.

6 Source-backed mechanisms

Table 4: Source and runtime provenance. Hashes are abbreviated here and complete in artifact manifests.

Study	file	SHA-256	audit use
Beauty Contest	game.py:119–124	exttt88eebe98a6c4...	returns the first legal integer in the entire response
Iterated PD	game.py:127–146	exttt88eebe98a6c4...	returns the earliest displayed cooperation/defection label
Observed Norms	norms.py:110–120	extttaf9c178887df...	parses the first in-range integer; all released outputs are exact integers
Persona Expression	README + world + sim	extttad426c187f11...	measurement description says one post per agent-step while tool.calling.mode is multi
Persona Expression	action_prompts.py + resolve.py	extttcc2236f2c2b6...	prompt permits batches and resolver executes every normalized call
Persona Expression	turns.py + SiliSocS 0.2.0	exttt8ad76033a969...	pins the one-agent-step policy and exact runtime wheel
Polarization	README + sim	exttt4b6117ec908e...	advertises up to three calls and configures max_actions: 3
Polarization	action_prompts.py + resolve.py	extttcc2236f2c2b6...	an entire multi-call batch is resolved before control returns
Polarization	turns.py + SiliSocS 0.2.0	exttt8ad76033a969...	the open-ended counter is updated after batch resolution

The released Beauty parser scans the full response and returns the first standalone integer in [11, 20]. The released IPD parser searches both displayed labels and returns whichever occurs earlier. Persona Expression combines a one-post-per-agent-step measurement description with multi-tool prompting and a resolver that executes every normalized call. Polarization’s open-ended policy updates its action counter only after an entire batch has been resolved; therefore a single response can cross the cap before the stopping condition is reevaluated. These are source-replayed statements for the pinned files above, not inferred latent mechanisms.

7 Input provenance

Study	official tarball	bytes	SHA-256
Beauty Contest	extttbeauty_contest_sweep.tar.gz	4,246,902	exttt62e80580beaa...
Iterated PD	extttiterated_pd_sweep.tar.gz	37,475,842	exttt8a5590c95b8a...
Persona Expression	extttpersona_expression_sweep.tar.gz	53,728,040	exttt6e8d28894d7d...
Observed Norms	extttobserved_norms_sweep.tar.gz	3,605,649	exttt686e324ae66b...
Polarization	extttpolarization_sweep.tar.gz	371,252,827	extttb506d9e84d76...

All analyses read compressed JSONL members directly from the official tarballs. This avoids Windows long-path truncation and records the exact tar member in run- or action-level tables. Manifests, tarballs, parser files, runtime wheel, generated artifacts, and paper files are hashed in machine-readable manifests.

8 Interpretive limits

1. Intervals are conditional on the released seeds and sweep support. They do not quantify uncertainty over providers, future model versions, or an open-ended population of prompts.
2. An execution discrepancy is not automatically a changed scientific verdict. The projection tables report when it matters for the registered contrast.
3. We do not rerun counterfactual societies after replacing parsers, truncating batches, or restoring rejected calls. Such interventions would change subsequent histories and require new simulations.
4. Persona lexical diversity is a transparent token-set Jaccard metric used for execution sensitivity, not a universal measure of persona expression quality.
5. Polarization edge alignment is a direct network-neighbor stance-similarity statistic derived from the released graph and opinion probes. It is not a causal estimate of exposure policy.
6. Human-anchor claims in Observed Norms remain U because the licensed WVS reference data are not distributed. No surrogate anchor is invented.

9 Reproduction contract

The public entry point runs unit tests, regenerates all five study artifact directories, rebuilds the cross-study summary, compiles the two-page paper and this supplement, validates the claim–evidence ledger, and checks page limits. Each study script fails on missing tar members, parser replay failure, malformed structured outputs, or unmatched claim arms rather than silently skipping them. Polarization is the exception only in the scientifically appropriate sense: failed probes and rejected calls are retained as explicit data-quality fields, while affected claim contrasts drop only rows whose required metric is undefined.

References

- [1] D. Ju, A. Williams, B. Karrer, and M. Nickel. Sense and sensitivity: Evaluating the simulation of social dynamics via LLMs. arXiv:2412.05093, 2024.
- [2] M. Linde, J. Sun, P. Balluff, D. Radovanović, and C.-h. Chan. Making uncertainty visible: Multiverse analysis for robust computational social science. arXiv:2605.19745, 2026.
- [3] S. Sarangi, M. P. Touzel, A. Bück-Kaeffer, Z. Yang, J.-F. Godbout, and R. Rabbany. EASE configuration facilitates a reproducible science of LLM social simulations. arXiv:2605.30258, 2026.
- [4] J. Ye, L. Cao, D. Chen, and E. Ferrara. Stop drawing scientific claims from LLM social simulations without robustness audits. arXiv:2605.18890, 2026.
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